2.

Electron rest mass , photon frequency ; assume it absorbs the photon completely and moves at ; momentum conservation then gives

The total energy after collision is

Eliminate using :

Electron - photon total energy before collision:

Using ,

This violates energy conservation, so the supposed process cannot occur.

Metal-bound electron + incident photon: momentum is not conserved in photoemission. Use energy conversion.

Escape from metal needs at least work (work function). After escape: speed , photon energy , electron rest mass . Photoemission requires

The emitted speed is much less than light speed; hence

Drop higher terms; substitute the first two of in :

Thus suffices for and photoemission.

3. For ease, consider the small block on .

as shown; are , acceleration magnitudes. , : block - wedge and wedge - table normal forces.

Let denote the wedge acceleration magnitude. Block equations:

Wedge equations:

The block is constrained to the slope: